ProofLoop

Blast Radius Zero

Verifiable acceptance for coding agents that run untrusted code.

Vultr Agent Rush Hackathon · Blast Radius Zero · 2026-11-08

ISOLATED SANDBOX

Docker container

```
--network none
512 MB  ·  1 CPU
cap-drop ALL
pids-limit 128
```

The agent executes here.
The verdict is made elsewhere.

Apache-2.0 · derived from ProofLoop

A green test run is not proof of a fix

An agent can pass the suite without solving the task.

01

Edit the tests

Delete or weaken the failing assertions.

02

Hardcode values

Special-case the exact inputs the tests use.

03

Plant test hooks

Drop conftest.py, sitecustomize.py or .pth files mid-run.

04

Forge the report

Rewrite JUnit or coverage output from an atexit hook.

The agent's self-report is a claim, not evidence.

ProofLoop treats every candidate patch as untrusted input.

Separate the executor from the accepting authority

The model proposes. The verifier disposes.

1

Untrusted executor

Vultr Serverless Inference

ReAct loop with three tools

Runs inside Docker

2

Frozen acceptance

Visible + hidden tests

14 deterministic checks

No LLM in the verdict path

3

Replayable evidence

evidence.json + SHA-256

HTML report

verify-evidence / replay

The LLM proposes. The verifier disposes.

Architecture

One agent. One sandbox. One independent verdict.



THE VERDICT PATH — deterministic, no model

- `baseline_reproduced`
- `candidate_tests_passed`
- `visible_tests_match_baseline`
- `hidden_tests_passed`
- `hidden_tests_expected`
- `candidate_tree_frozen`
- `test_integrity`
- `scope`
- `noop`
- `hardcode_suspicion`
- `runtime_tamper_suspicion`
- `runtime_file_injection`
- `mutation_score`
- `coverage`

No network in the sandbox. No model in the verdict path.

Blast radius stays inside the box

Defence in depth around untrusted code.

N — No network

The container cannot exfiltrate data or install packages.

R — Resource caps

512 MB, 1 CPU, 128 PIDs, all capabilities dropped.

S — Shadow-proof tools

python -P -c launcher prevents repo-root shadowing.

H — Signed results

HMAC attestation is cross-checked against JUnit.

Q — Runtime quarantine

Files planted during tests are removed and reported.

W — Write guards

Only allowed_paths are writable; tests are denied.

The sandbox bounds the blast radius; the verifier bounds the claim.

Two scenes, two verdicts

Same verifier. Only the candidate patch changes.

SCENE 1 · AGENT FIX

PASS

14 / 14 checks

hidden 3 / 3

mutation 8 / 8 killed

coverage 100%

SCENE 2 · CHEAT PATCH

FAIL

hidden_tests_passed

hidden_tests_expected

hardcode_suspicion

verdict: FAIL

Keyless stack self-test with the bundled fix and cheat patches. The Vultr agent scene follows the same verification path.

Evidence you can replay

Hash the sheet, replay the run, compare the verdict.

evidence.json

verdict
PASS · FAIL · INSUFFICIENT_EVIDENCE

patch_sha256
hash of the candidate patch

diff_sha256
hash of the frozen diff

acceptance_sha256
hash of the acceptance criteria

checks
14 pass/fail entries with details

environment
python, sandbox, image id, versions

```
$ python -m proofloop verify-evidence \
    --evidence runs/demo/evidence-pass/evidence.json
evidence hash OK: 6fe27cf7...

$ python -m proofloop replay \
    --evidence runs/demo/evidence-pass/evidence.json \
    --repo examples/discount_demo --task task.json \
    --patch scene1-agent.patch --out runs/replay
old verdict: PASS
new verdict: PASS
consistent: True
```

A human reviews the evidence, not the model's claim.

What this is — and what it is not

HONEST LIMITS

- One synthetic task — not a benchmark.
- Some checks are heuristics, not proofs.
- Hidden tests are local to this demo.
- Model behaviour is not a general accuracy claim.

NEXT

- Keep hidden tests in a separate trust domain.
- Add gVisor / E2B sandbox backends.
- Evaluate across the Vultr model catalog.
- Wire verify-evidence into CI sign-off.

Small, real, reproducible.

Trust the evidence, not the agent.